

Industry-specific AI solution.

AI Capstone

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Project Overview

This project presents an **integrated AI system for healthcare triage**, designed to support outpatient diabetes follow-up prioritization. The system addresses a real-world problem: clinics often lack the time and resources to proactively identify patients whose condition may be worsening between visits.

The solution combines multiple AI domains into a **cohesive workflow**, including:

- statistical data analysis
- machine learning prediction
- generative AI explanations
- agentic decision routing

Rather than focusing on a single model, the project emphasizes **system-level design, ethical reasoning, and professional applicability**, aligning with industry expectations for AI deployment.

Dataset Description

The dataset used in this project is **synthetic and privacy-safe**, simulating outpatient diabetes patient records.

Key features include:

- A1C levels
- systolic blood pressure
- missed appointments
- medication adherence
- emergency room visits
- patient demographics

The dataset represents **risk indicators commonly used in clinical decision-making**, allowing the system to model realistic patient prioritization scenarios without exposing sensitive health data.

Data Preparation & Exploration

Data preparation involved:

- cleaning and structuring patient variables
- validating ranges and distributions

- identifying relevant predictors of risk

Exploratory Data Analysis (EDA) was used to:

- analyze distributions of A1C and blood pressure
- identify correlations between adherence and outcomes
- detect patterns in missed appointments and ER visits

This step ensured that the model inputs were **interpretable, clinically meaningful, and aligned with real-world signals**, rather than relying on arbitrary features.

Model Design

The system uses a **logistic-style supervised learning model** to estimate the probability that a patient requires near-term intervention.

Key design choices:

- **Interpretable model (logistic classifier)** instead of black-box models
- Feature set based on clinically relevant indicators
- Output: probability score → mapped into risk tiers

The model is embedded within a **multi-layer architecture** that includes:

1. Data profiling
2. Risk prediction
3. Generative explanation
4. Agentic routing

This design transforms predictions into **actionable decisions**, not just scores.

Training Process

The training process followed standard machine learning practices:

- dataset split into training and testing sets
- feature selection based on domain relevance
- model fitting using supervised learning
- evaluation on unseen test data

Because the dataset is synthetic, the focus was not on optimizing performance for deployment, but on:

- demonstrating **correct methodology**
- ensuring **logical consistency**
- validating **end-to-end pipeline behavior**

Model Evaluation

The model was evaluated using standard classification metrics:

- **Accuracy:** 0.85
- **ROC AUC:** 0.888
- **Positive rate:** 0.367

Evaluation also included:

- risk tier distribution
- proportion of high-risk patients
- system-level behavior (routing outcomes)

The evaluation demonstrates that the system produces **consistent and usable prioritization signals**, even though it is not clinically validated.

Results & Interpretation

The system successfully:

- identified high-risk patients
- categorized patients into actionable tiers
- generated readable summaries
- routed cases to appropriate follow-up actions

Key insight:

The value of the system is not just prediction accuracy, but **its ability to integrate prediction with explanation and action**.

This demonstrates that **end-to-end workflow design** is more impactful than isolated model performance.

Non-Technical Explanation

This system helps clinics decide **which patients need attention first**.

Instead of reviewing every patient manually, the system:

1. Looks at important health signals
2. Calculates a risk level
3. Explains why the patient is at risk
4. Suggests what to do next

For example:

- Low risk → reminder message
- Medium risk → nurse follow-up
- High risk → doctor review

The system **does not replace doctors**. It simply helps them focus on the patients who need help the most.

Experimental Design Justification

The experimental design was guided by:

- real-world healthcare constraints
- need for interpretability
- ethical considerations

Key decisions:

- Use of **synthetic data** → protects privacy
- Use of **interpretable model** → supports trust
- Inclusion of **human-in-the-loop** → ensures safety

This design prioritizes:

- reliability over complexity
- transparency over optimization
- safety over automation

Workflow Completeness

The system demonstrates a **complete AI workflow**, including:

- data ingestion and preprocessing
- exploratory analysis
- model training and evaluation
- output generation
- decision routing
- documentation and reproducibility

It also includes:

- architecture design
- notebook and executable script
- outputs and metrics
- presentation and synthesis paper

This ensures the project is **end-to-end and production-aware**, even as a prototype.

Bias and Risk Awareness

Several risks were identified:

- **Bias risk:** historical healthcare disparities
- **False negatives:** missing high-risk patients
- **Automation bias:** over-reliance on AI
- **Privacy concerns:** sensitive patient data

Mitigation strategies:

- use of transparent features
- human review for high-risk cases
- clear system limitations
- decision-support (not decision-making) design

These align with responsible AI frameworks such as:

- NIST AI RMF
- WHO AI ethics guidelines

Future Improvements & Integration

Future enhancements include:

- validation with real-world clinical datasets
- fairness and subgroup performance testing
- integration with electronic health records (EHR)
- longitudinal patient tracking
- monitoring dashboards and feedback loops

Additionally, the system could evolve into a **production-ready clinical decision support tool** with proper governance and validation.

Conclusion

This project demonstrates how an AI system can:

- integrate multiple domains into a unified workflow
- support real-world decision-making
- balance performance with ethical responsibility

More importantly, it shows that **modern AI work is about system design, not just models**.

The project reflects **professional readiness** by combining:

- technical skills
- ethical awareness
- practical application

Reproducibility

To reproduce the project:

```
Start by cloning this Repo: https://github.com/atamayop2018/integrative-industry-synthesis  
python -m venv .venv  
source .venv/bin/activate  
pip install -r requirements.txt  
python src/healthcare_triage_ai.py
```

Additional options:

- Run the Jupyter notebook version

- Review outputs in `/outputs` folder

The `requirements.txt` file ensures consistent environment setup.

References

- National Institute of Standards and Technology (2023). *AI Risk Management Framework (AI RMF 1.0)*
- World Health Organization (2021). *Ethics and Governance of AI for Health*
- Obermeyer et al. (2019). *Dissecting racial bias in healthcare algorithms*
- Topol, E. (2019). *High-performance medicine*
- Udacity AI Nanodegree Course Materials